

# ***Folate intake, MTHFR polymorphisms, and risk of esophageal, gastric, and pancreatic cancer: a meta-analysis.***

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**BACKGROUND & AIMS:** Increasing evidence suggests that a low folate intake and impaired folate metabolism may be implicated in the development of gastrointestinal cancers. We conducted a systematic review with meta-analysis of epidemiologic studies evaluating the association of folate intake or genetic polymorphisms in 5,10-methylenetetrahydrofolate reductase (MTHFR), a central enzyme in folate metabolism, with risk of esophageal, gastric, or pancreatic cancer. **METHODS:** A literature search was performed using MEDLINE for studies published through March 2006. Study-specific relative risks were weighted by the inverse of their variance to obtain random-effects summary estimates. **RESULTS:** The summary relative risks for the highest versus the lowest category of dietary folate intake were 0.66 (95% confidence interval [CI], 0.53-0.83) for esophageal squamous cell carcinoma (4 case-control), 0.50 (95% CI, 0.39-0.65) for esophageal adenocarcinoma (3 case-control), and 0.49 (95% CI, 0.35-0.67) for pancreatic cancer (1 case-control, 4 cohort); there was no heterogeneity among studies. Results on dietary folate intake and risk of gastric cancer (9 case-control, 2 cohort) were inconsistent. In most studies, the MTHFR 677TT (variant) genotype, which is associated with reduced enzyme activity, was associated with an increased risk of esophageal squamous cell carcinoma, gastric cardia adenocarcinoma, noncardia gastric cancer, gastric cancer (all subsites), and pancreatic cancer; all but one of 22 odds ratios were >1, of which 13 estimates were statistically significant. Studies of the MTHFR A1298C polymorphism were limited and inconsistent. **CONCLUSIONS:** These findings support the hypothesis that folate may play a role in carcinogenesis of the esophagus, stomach, and pancreas.